INCH-POUND

MIL-DTL-85069E
15 November 2023
SUPERSEDING
MIL-DTL-85069D
See 6.3

DETAIL SPECIFICATION

SLEEVES AND GROMMETS, FASTENER

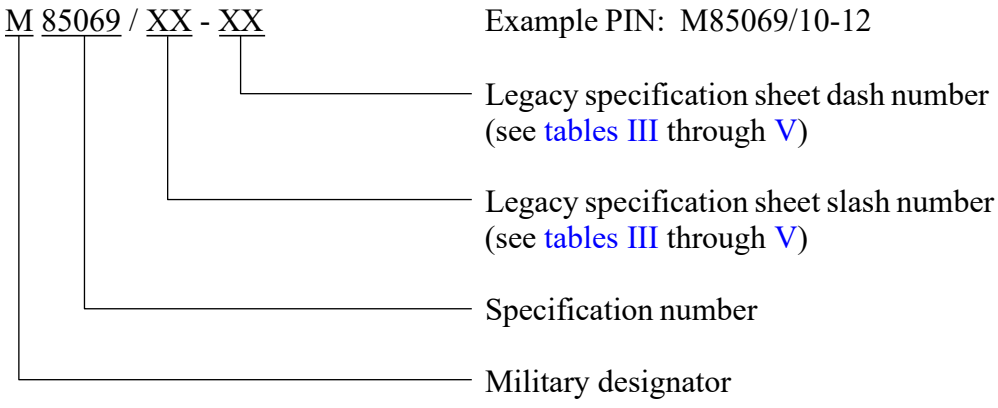
This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fastener sleeves and grommets for use in over-sized fastener holes in aerospace structures.

1.2 Classification. Sleeves and grommets are classified as shown in tables III through V.

1.3 Part or identification number (PIN). The PIN to be used for sleeves and grommets acquired to this specification is created as follows (see 6.2 and tables III through V). Part numbers from the legacy specification sheets have been retained to support continued procurement.



Comments, suggestions, or questions on this document should be addressed to DLA Aviation VEB, 6090 Strathmore Rd, Richmond, VA 23237, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASME

ASME B46.1	- Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available online at http://www.asme.org/

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	- Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online from http://www.asq.org/

SAE INTERNATIONAL

SAE AMS2700	- Passivation of Corrosion Resistant Steels
SAE AMS 5525	- Steel, Corrosion and Heat-Resistant, Sheet, Strip, and Plate 15Cr - 25.5Ni - 1.2Mo - 2.1Ti - 0.006B - 0.30V 1800 °F (982 °C) Solution Heat Treated
SAE AS87132	- Lubricant, Cetyl Alcohol, 1-Hexadecanol, Application to Fasteners

(Copies of these documents are available online at http://www.sae.org

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Conformance inspection. When specified (see 6.2), samples shall be subjected to conformance inspection in accordance with 4.2.

3.2 Design and construction.

3.2.1 Construction. The fastener sleeve or grommet shall be drawn from sheet or strip stock to form a one-piece tubular section and head. The thickness of the tubular section shall conform to the relevant figure. Sleeve tubular sections shall be grooved in 0.062-inch increments to aid in adjusting the sleeve to the desired length.

3.2.2 Material. Sleeve and grommet material shall conform to UNS S66286 corrosion and heat-resistant steel in accordance with SAE AMS 5525 (see 4.3.3).

3.2.3 Head structure and grain flow. When visually examined, the sleeve or grommet may show axial grain flow lines in the tubular section typical of a drawn sleeve or grommet. The flow lines shall be continuous at the fillet radius.

3.2.4 Protective coatings. The fastener sleeves and grommets shall be passivated in accordance with SAE AMS2700.

3.2.5 Lubrication. The fastener sleeves and grommets shall be lubricated with Cetyl alcohol or left bare. Cetyl alcohol, if used, shall conform to SAE AS87132. Packaging shall be marked as shown to indicate use or non-use, as indicated:

L - lubricated with Cetyl alcohol
B - no lubrication

3.2.6 Surface texture. Surfaces shall be free of cracks or draw marks; slight visible orange peeling effect shall be permitted. Unless otherwise specified (see 6.2), grommet surface texture shall be not more than 125 Ra microinches in accordance with ASME B46.1.

3.2.7 Shape. Fastener sleeves and grommets shall conform relevant figure and shall show no evidence of distortion.

3.2.8 Grip length. Grooved sleeves or non-grooved grommets may be procured at the required length. Alternatively, grooved sleeves may be broken at incremental lengths using the proper tooling . The step plate tool indicates the grip length at which the sleeve will break along the groove without damage to the sleeve.

3.3 Metallurgical properties.

3.3.1 Cracks. The fastener sleeves and grommets shall be free of cracks. A crack is identified as a clean crystalline break passing through the grain or boundary without the inclusion of foreign elements.

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3.3.2 Discontinuities. There shall be no laps or folds, seams or discontinuities, or other interruption in grain flow.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as conformance inspection (see 4.2).

4.2 Conformance inspection.

4.2.1 Sampling for examination. Test samples shall be selected at random in accordance with ASQ Z1.4, with the sampling level as specified in table I. The acceptance quality limit (AQL) shall be as specified in the contract or purchasing order (see 6.2). Identical sample items may be used for any of the tests, provided that selection of random samples is maintained and known characteristics of the samples are not used to influence the integrity of the test results.

TABLE I. Conformance examination.

Test	Reference paragraph	ASQ Z1.4 sampling inspection level
Inspection lot	4.2.2	-
Examination	4.3.1	II
Material certification	4.3.3	-
Grip length break	4.3.4	II

4.2.2 Inspection lot. A lot shall consist of sleeves or grommets made from the same lot of material and fabricated as a single continuous production run in which the forming machine set-up remains unchanged except for head configuration. A lot shall be restricted to one sleeve or grommet diameter or material thickness. A lot of material shall be defined as the quantity of sheet or strip steel made from a single heat. It shall be controlled and shipped with appropriate mill certification. In addition, the manufacturer shall maintain a record of inspections by lot number for a period of three years.

4.2.3 Certified test report. The manufacturer shall furnish certification that the product conforms to this specification. The record shall state that the product meets all the requirements of this specification.

4.3 Examination and tests.

4.3.1 Examination. Each of the sample sleeves or grommets selected for examination shall be examined for conformance to the requirements for dimensions, workmanship, identification, and to all requirements not covered by the tests. Samples failing to meet examination requirements shall result in rejection of the lot.

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4.3.2 Strength. The strength of the fastener sleeve or grommet combination is considered the strength of the fastener alone. The wall thickness of the sleeve or grommet is considered thin enough so that there shall be no material decrease in thickness when under load. Sleeve or grommet thickness shall be measured by any approved means. Sleeves or grommets that do not meet the dimensional requirements for thickness as shown in figures 1 through 4 shall result in rejection of the lot.

4.3.3 Material certification. Mill certification on all materials shall be kept on file by the manufacturer and shall be available to the procuring agency for a period of three years.

4.3.4 Grip length break. Using different fastener sleeves, break at different lengths with the proper tooling. Grooved sleeves shall be broken only one time. To adjust to length at a location other than the breaking area of the step plate, the sleeve may be ground to length using the painted section of the step plate. Sleeves that fail to break cleanly at the grooves shall result in rejection of the lot.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fastener sleeves and grommets specified herein are intended for structural and non-structural application in aerospace structures and other military equipment. The function of the sleeves and grommets is to serve as a repair method for holes that have become damaged, elongated, or corroded, and to protect the structure in soft or brittle material. The sleeve is intended to increase the integrity of the overall joint, not the strength of the individual fastener.

6.1.1 Length adjustment. Fastener sleeves and grommets that are manufactured with proper head configuration and diameter may be procured at the required length. Alternatively, sleeves may be adjusted to the correct length for each individual fastener.

6.1.2 Installation of sleeves and grommets. Fastener sleeves and grommets should be installed with reasonable care to ensure that they seal properly, inhibit corrosion, and establish, preserve, or restore the structural integrity of joints in modification, repair, or in the original manufacture of aerospace or military hardware. The installation information and the processing

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specifications should be followed. Tools are available to facilitate and ensure proper installation of the sleeve or grommet.

6.1.2.1 General purpose and restrictions. The fastener sleeves and grommets will be used to replace oversize shank fasteners for repair and maintenance. The sleeves and grommets will be used with standard diameter straight shank fasteners. They will be used under the same restrictions as oversize fasteners and in compliance with the specific user's repair procedures and restrictions.

6.1.2.2 Number of fasteners. There will be no restrictions on the number of fasteners installed with sleeves or grommets when the edge distance from the center of the hole is two times the diameter and greater.

6.1.2.3 Fasteners. The fastener or attaching element (nut or collar) will contain a counter-bore to accept the protrusion of the sleeve or grommet; or a washer will be used; or the sleeve or grommet will be flush on the backside of the stack.

6.1.2.4 Fastener restrictions. The fastener sleeves or grommets will not be used with tapered shank fasteners or blind fasteners.

6.1.2.5. Interference. Interference (hole-to-sleeve) will be not greater than 0.003 inch where the material thickness is less than 0.5 inch. Interference will be not greater than 0.0015 inch where material thickness is greater than 0.5 inch.

6.1.3 Applications. Appropriate engineering approval will be required in the following applications:

a. In fluid containers having no self-sealing feature, as in fuel, oil, or water tanks.
b. Fatigue and fracture critical locations.
c. Where the edge distance from the center of the hole is less than 1.5 diameters.
d. Where the number of discrepant holes being repaired is greater than 10-percent of the total number of holes in a pattern.
e. Where more than three adjacent holes are discrepant.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. PIN to include classification of sleeve or grommet required (see 1.3).
c. Conformance inspection requirements (see 3.1).
d. Surface texture exceptions (see 3.2.6).
e. AQL (see 4.2.1).
f. Packaging (see 5.1).

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6.3 Supersession data. This specification supersedes the documents listed in table II.

TABLE II. Supersession data.

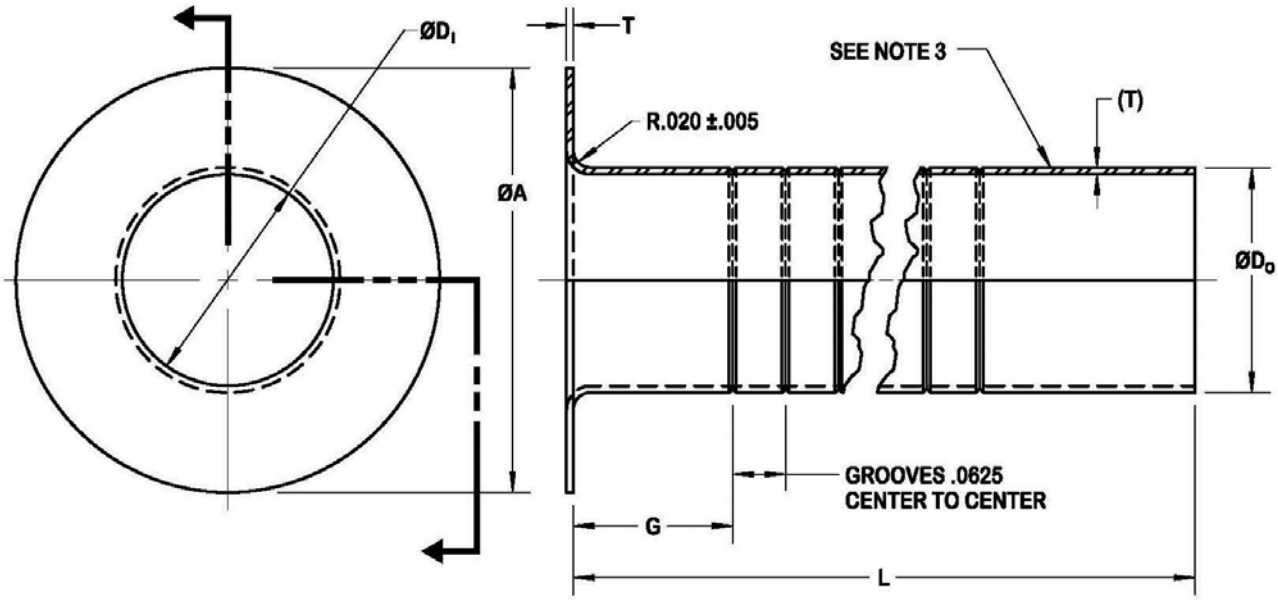
Document	Date
MIL-DTL-85069D	10 February 2012
MIL-S-85069/1A	16 September 1981
MIL-DTL-85069/2B	25 January 2006
MIL-DTL-85069/3B	25 January 2006
MIL-DTL-85069/4B	25 January 2006
MIL-DTL-85069/5B	25 January 2006
MIL-DTL-85069/6B	25 January 2005
MIL-DTL-85069/7B	25 January 2006
MIL-DTL-85069/8B	25 January 2006
MIL-DTL-85069/9B	25 January 2006
MIL-DTL-85069/10A	25 January 2006
MIL-DTL-85069/11A	25 January 2006

6.4 Subject term (key word) listing.

Aircraft
Cadmium
Countersunk
Discrepant hole
Drawn
Flush
Grip length
Grooved
Hard seat
One piece
Oversize
Protection
Protruding head
Reduced head
Shank
Shear
Step plate
Structural
Swaged
Tension
Tubular

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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NOTES:
1. Dimensions are in inches and shall apply prior to surface finish. Refer to table III.
2. Sleeves may be procured in standard required lengths or may be adjusted to required length as needed.
3. Sleeves 05 and 55, ungrooved for last 0.188 Ref of L.
   Sleeves 06 and 08, ungrooved for last 0.250 Ref of L.
   Sleeves 10 and 12, ungrooved for last 0.312 Ref of L.
4. Tolerance 0.010 unless otherwise marked.

FIGURE 1. Grooved sleeve fastener, protruding head.

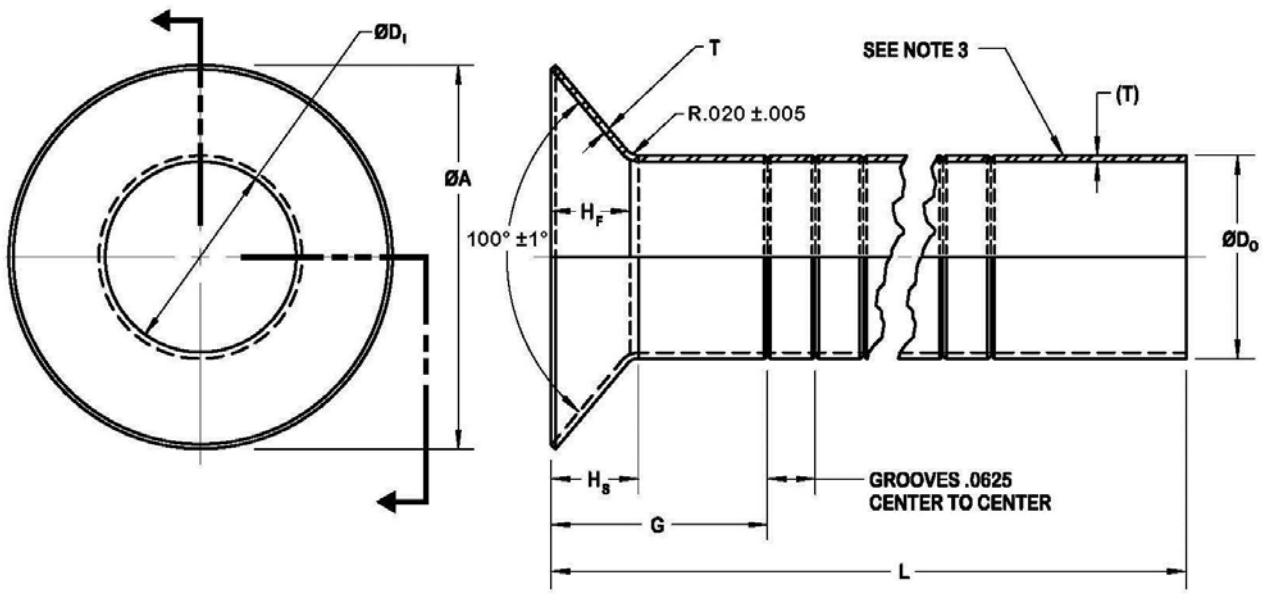
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TABLE III. Grooved sleeve fastener, protruding head - part numbers and dimensions.

MS part number [1]	MS dash number	Fastener size	ØA ±0.005	$ØD_I$ max. min.	$ØD_O$ max. min.	G Ref	L ±0.010	T +0.002 −0.003	Drill number [2] (diameter)
Protruding head, shear and tension, 1/64 oversize									
M85069/2	05	5/32	0.312	0.1565 0.1550	0.1730 0.1720	0.125	0.750	0.008	17 (0.1730)
	55	#8	0.325	0.1635 0.1620	0.1800 0.1790	0.125	0.750	0.008	15 (0.1800)
	06	#10	0.380	0.1895 0.1880	0.2055 0.2045	0.125	1.00	0.008	5 (0.2055)
	08	1/4	0.500	0.2495 0.2480	0.2656 0.2646	0.188	1.00	0.008	17/64 (0.2656)
	10	5/16	0.620	0.3120 0.3105	0.3281 0.3271	0.250	1.00	0.008	21/64 (0.3281)
	12	3/8	0.750	0.3745 0.3730	0.3906 0.3896	0.312	1.00	0.008	25/64 (0.3906)
MS part number [1]	MS dash number	Fastener size	ØA ±0.005	$ØD_I$ max. min.	$ØD_O$ max. min.	G	L ±0.010	T ±0.001	Drill number [2] (diameter)
Protruding head, shear and tension, 1/32 oversize									
M85069/6	06	#10	0.380	0.1895 0.1880	0.2210 0.2200	0.125	1.00	0.015	2 (0.2210)
	08	1/4	0.500	0.2495 0.2480	0.2812 0.2802	0.188	1.00	0.015	9/32 (0.2812)
	10	5/16	0.620	0.3120 0.3105	0.3437 0.3427	0.250	1.00	0.015	11/32 (0.3437)
	12	3/8	0.750	0.3745 0.3730	0.4062 0.4052	0.312	1.00	0.015	13/32 (0.4062)

[1] MS part numbers retained for legacy support purposes.
[2] Recommended for metal-to-metal clearance fit of sleeve to hole.
3 Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.

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NOTES:
1. Dimensions are in inches and shall apply prior to surface finish. Refer to table IV.
2. Sleeves may be procured in standard required lengths or may be adjusted to required length as needed.
3. Sleeves 05 and 55, ungrooved for last 0.188 Ref of L.
   Sleeves 06 and 08, ungrooved for last 0.250 Ref of L.
   Sleeves 10 and 12, ungrooved for last 0.312 Ref of L.
4. Tolerance 0.010 unless otherwise marked.
5. 100-degree angle can be measure on internal or external surface of sleeve.

FIGURE 2. Grooved sleeve fastener, 100-degree head.

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TABLE IV. Grooved sleeve fastener, 100-degree head - part numbers and dimensions.

MS part number [1]	MS dash number	Fastener size	ØA ±0.005	$ØD_I$ max. min.	$ØD_O$ max. min.	G Ref	H_F Ref	H_S Ref	L ±0.010	T +0.002 −0.003	Drill number [2] (diameter)
Flush reduced head, shear, 1/64 oversize											
M85069/3	05	5/32	0.252	0.1565 0.1550	0.1730 0.1720	0.125	0.040	0.045	0.750	0.008	17 (0.1730)
	55	#8	0.260	0.1635 0.1620	0.1800 0.1790	0.125	0.041	0.046	0.750	0.008	15 (0.1800)
	06	#10	0.300	0.1895 0.1880	0.2055 0.2045	0.125	0.047	0.052	1.00	0.008	5 (0.2055)
	08	1/4	0.393	0.2495 0.2480	0.2656 0.2646	0.188	0.061	0.066	1.00	0.008	17/64 (0.2656)
	10	5/16	0.472	0.3120 0.3105	0.3281 0.3271	0.250	0.068	0.073	1.00	0.008	21/64 (0.3281)
	12	3/8	0.558	0.3745 0.3730	0.3906 0.3896	0.312	0.078	0.083	1.00	0.008	25/64 (0.3906)
Flush head, 1/64 oversize											
M85069/4	05	5/32	0.282	0.1565 0.1550	0.1730 0.1720	0.125	0.053	0.058	0.750	0.008	17 (0.1730)
	55	#8	0.294	0.1635 0.1620	0.1800 0.1790	0.125	0.055	0.060	0.750	0.008	15 (0.1800)
	06	#10	0.347	0.1895 0.1880	0.2055 0.2045	0.125	0.071	0.076	1.00	0.008	5 (0.2055)
	08	1/4	0.470	0.2495 0.2480	0.2656 0.2646	0.188	0.098	0.103	1.00	0.008	17/64 (0.2656)
	10	5/16	0.558	0.3120 0.3105	0.3281 0.3271	0.250	0.109	0.114	1.00	0.008	21/64 (0.3281)
	12	3/8	0.688	0.3745 0.3730	0.3906 0.3896	0.312	0.137	0.142	1.00	0.008	25/64 (0.3906)
Flush full head, tension, 1/64 oversize											
M85069/5	05	5/32	0.312	0.1565 0.1550	0.1730 0.1720	0.125	0.065	0.070	0.750	0.008	17 (0.1730)
	55	#8	0.328	0.1635 0.1620	0.1800 0.1790	0.125	0.070	0.075	0.750	0.008	15 (0.1800)
	06	#10	0.380	0.1895 0.1880	0.2055 0.2045	0.125	0.080	0.085	1.00	0.008	5 (0.2055)
	08	1/4	0.505	0.2495 0.2480	0.2656 0.2646	0.188	0.108	0.113	1.00	0.008	17/64 (0.2656)
	10	5/16	0.631	0.3120 0.3105	0.3281 0.3271	0.250	0.135	0.140	1.00	0.008	21/64 (0.3281)
	12	3/8	0.758	0.3745 0.3730	0.3906 0.3896	0.312	0.162	0.167	1.00	0.008	25/64 (0.3906)

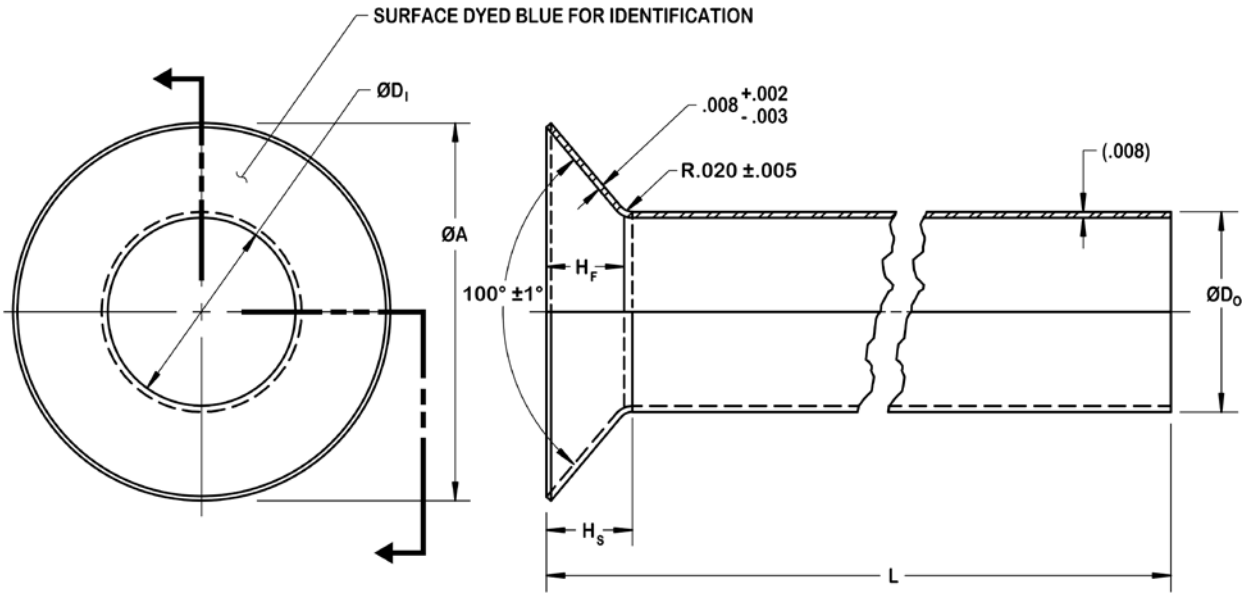
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TABLE IV. Grooved sleeve fastener, 100-degree head - part numbers and dimensions - Continued.

MS part number [1]	MS dash number	Fastener size	ØA ±0.005	$ØD_I$ max. min.	$ØD_O$ max. min.	G Ref	H_F Ref	H_S Ref	L ±0.010	T +0.002 −0.003	Drill number [2] (diameter)
Flush reduced head, shear, 1/32 oversize											
M85069/7	06	#10	0.300	0.1895 0.1880	0.2210 0.2200	0.125	0.047	0.057	1.00	0.016	2 (0.2210)
	08	1/4	0.393	0.2495 0.2480	0.2812 0.2802	0.188	0.061	0.071	1.00	0.016	9/32 (0.2812)
	10	5/16	0.472	0.3120 0.3105	0.3437 0.3427	0.250	0.068	0.078	1.00	0.016	11/32 (0.3437)
	12	3/8	0.558	0.3745 0.3730	0.4062 0.4052	0.312	0.078	0.088	1.00	0.016	13/32 (0.4062)
Flush head, 1/32 oversize											
M85069/8	06	3/16	0.347	0.1895 0.1880	0.2210 0.2200	0.125	0.071	0.081	1.00	0.016	2 (0.2210)
	08	1/4	0.470	0.2495 0.2480	0.2812 0.2802	0.188	0.098	0.108	1.00	0.016	9/32 (0.2812)
	10	5/16	0.558	0.3120 0.3105	0.3437 0.3427	0.250	0.109	0.119	1.00	0.016	11/32 (0.3437)
	12	3/8	0.688	0.3745 0.3730	0.4062 0.4052	0.312	0.137	0.147	1.00	0.016	13/32 (0.4062)
Flush full head, tension, 1/32 oversize											
M85069/9	06	#10	0.380	0.1895 0.1880	0.2210 0.2200	0.125	0.080	0.090	1.00	0.016	2 (0.2210)
	08	1/4	0.505	0.2495 0.2480	0.2812 0.2802	0.188	0.108	0.118	1.00	0.016	9/32 (0.2812)
	10	5/16	0.631	0.3120 0.3105	0.3437 0.3427	0.250	0.135	0.145	1.00	0.016	11/32 (0.3437)
	12	3/8	0.758	0.3745 0.3730	0.4062 0.4052	0.312	0.162	0.172	1.00	0.016	13/32 (0.4062)

[1] MS part numbers retained for legacy support purposes.
[2] Recommended for metal-to-metal clearance fit of sleeve to hole.
3. Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.

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NOTES:
1. Dimensions are in inches and shall apply prior to surface finish. Refer to table V.
2. Grommets may be procured in standard required lengths or may be adjusted to required length as needed.
3. Tolerance 0.010 unless otherwise marked.
4. 100-degree angle can be measure on internal or external surface of sleeve.

FIGURE 3. Grommet fastener.

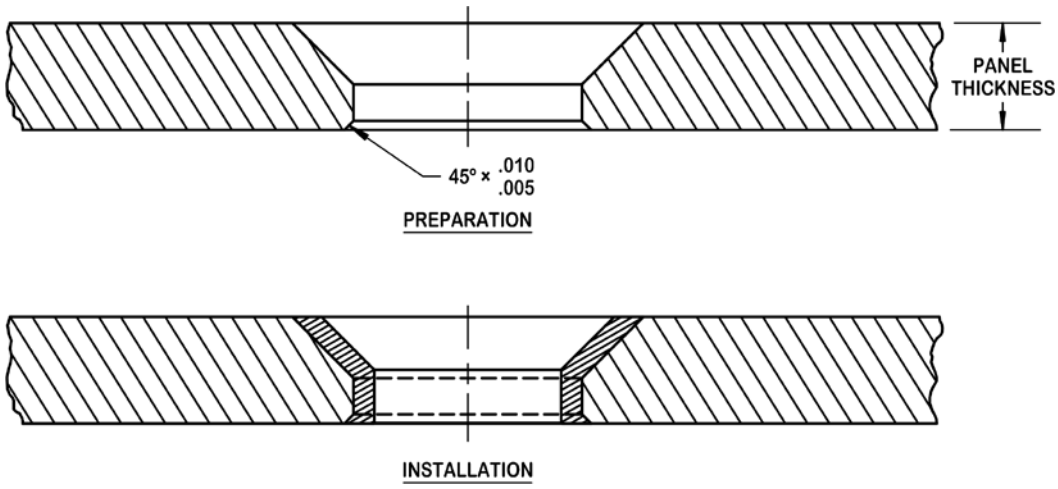
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TABLE V. Grommet fastener - part numbers and dimensions.

MS part number [1]	MS dash number	Fastener size	ØA ±0.005	$ØD_I$ max. min.	$ØD_O$ max. min.	H_F [2] Ref	H_S Ref	L [3] ±0.010	Drill diameter [4]
Reduced head, swaged, countersunk, panel fastener									
M85069/10	08	1/4	0.393	0.2545 0.2530	0.2706 0.2696	0.061	0.066	1.00	0.273
	10	5/16	0.472	0.3170 0.3155	0.3331 0.3321	0.068	0.073	1.00	0.334
	12	3/8	0.558	0.3795 0.3780	0.3956 0.3946	0.078	0.083	1.00	0.398
Tension head, swaged, countersunk, panel fastener									
M85069/11	08	1/4	0.505	0.2545 0.2530	0.2706 0.2696	0.108	0.113	1.00	0.273
	10	5/16	0.631	0.3170 0.3155	0.3331 0.3321	0.135	0.140	1.00	0.334
	12	3/8	0.758	0.3795 0.3780	0.3956 0.3946	0.162	0.167	1.00	0.398

[1] MS part numbers retained for legacy support purposes.
[2] Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.
[3] Uncut grommet length is 1.000 inch. Before installing, cut grommet length to equal panel thickness (see figure 4).
[4] Special drill is required.

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Check the source to verify that this is the current version before use.



NOTE:
1. Dimensions are in inches.

FIGURE 4. Grommet hole preparation and installation.

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Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - GS

Review Activity:
Air Force - 71

Preparing Activity:
DLA - GS4

(Project 5365-2024-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.dla.mil/

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